

Multi-Agent Communication Graph Generation via Score-Based Diffusion

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Abstract

In Multi-Agent Systems (MAS), unstructured aggregation of agents often induces a high-energy state—formulated as a composite of high communication costs and low reasoning utility—where redundant communication graphs drown out valuable signals between agents. Existing paradigms treat graph generation as a black-box search, overlooking the latent gradients of energy, thus yielding suboptimal communication graphs. In this paper, we bridge the energy landscape and generative modeling to formalize graph generation as an energy minimization process. This perspective transforms the challenge from discrete combinatorial search into a continuous navigation task. To instantiate this, we propose MAGE, a score-based diffusion framework that, by exploiting the duality between the generative score and the negative energy gradient, constructs task-adaptive graphs via iterative denoising driven by two-stage gradient guidance. Extensive experiments across six benchmarks demonstrate that MAGE achieves state-of-the-art performance. Visualization further confirms that MAGE successfully escapes high-energy states, generating rational graphs.

1. INTRODUCTION

Any self-organizing system that is at equilibrium with its environment must minimize its free energy (Friston, 2010; 2019). The principle, while rooted in thermodynamics, offers a powerful intuition for artificial intelligence: true collective intelligence emerges through the minimization of

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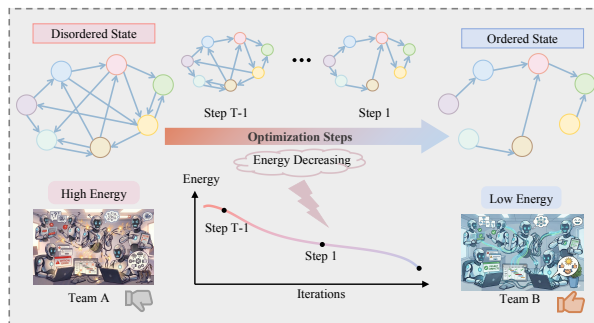


Figure 1. Illustration of agent team configuration within an energy landscape. Higher energy levels correspond to disordered graphs with redundant connections, while deep energy valleys represent streamlined, efficient topologies capable of effective task resolution. Distinct node colors denote specific agent roles (e.g., Math Solver). Arrows ($\rightarrow$) indicate workflow dependencies, where a successor conditions its actions on the predecessor’s output.

energy (e.g., friction and uncertainty). As the field transitions from monomeric Large Language Models (LLMs) to complex Multi-Agent Systems (MAS) (Zhuge et al., 2024; Zhang et al., 2025c;b), optimizing this emergence becomes critical. In a MAS, coordination is governed by a communication graph, where nodes represent agents and edges represent workflow dependencies. However, naively connecting these nodes typically results in a **high-energy state**: an over-connected and disordered graph where excessive edges spike token costs and unstructured noise drowns out valuable signals, ultimately degrading task utility. Therefore, to surpass the capabilities of individual agents (Zhang et al., 2024; Qian et al., 2025), the system must minimize its energy (manifested as a metric that unifies coordination cost and utility), converging from a disordered communication graph to an ordered one, as shown in Figure 1.

However, achieving this energy descent is non-trivial. Existing paradigms—ranging from pruning (Zhang et al., 2025b) and RL-based edge optimization (Zhuge et al., 2024; Zhang et al., 2025a) to autoregressive generation (Li et al., 2026) and LLM-driven planning (Zhang et al., 2025d; Hu et al., 2025)—face fundamental limitations. Critically, these methods overlook the latent energy landscape. For instance, some approaches are trained solely on a set of low-energy graphs

without access to their exact energy values (Zhang et al., 2025d; Hu et al., 2025; Li et al., 2026), while others (Zhuge et al., 2024; Zhang et al., 2025a) only derive energy values. By treating the generation process as a black box, they are forced to navigate the high-dimensional space blindly without gradient guidance, prone to becoming trapped in local optima and yielding metastable communication graphs.

In this paper, we unveil multi-agent communication graph generation as an inherent energy minimization process. Inspired by the thermodynamic intuition of self-organizing systems, we conceptualize the objective existence of MAS energy—manifested as the unification of coordination cost and utility (Zhang et al., 2025b)—into a scalar energy function. Under this conceptualization, the synthesis of a communication graph transforms from a search problem (Zhang et al., 2025a) into a continuous navigation task: the goal is to descend the naturally occurring energy landscape—a high-dimensional surface where valleys correspond to optimal collaboration graphs.

To instantiate this conceptualization, we propose **MAGE** (i.e., **M**ulti-**A**gent **C**ommunication **G**raph **G**eneration), a score-based diffusion framework engineered to traverse the energy landscape. Drawing an analogy with thermodynamics, where stable systems favor low-energy states, we model the distribution of optimal communication graphs as a Boltzmann distribution ($p \propto e^{-E}$). This allows us to exploit the fundamental mathematical duality between the generative score function ($\nabla \log p$) and the negative energy gradient ($-\nabla E$). Leveraging this duality, MAGE employs a two-stage mechanism to achieve energy minimization. First, the primary diffusion backbone acts as an implicit energy optimizer, utilizing a joint graph transformer to estimate the score and guide the system into low-energy manifolds. Second, to preclude convergence to generic sub-optima, we integrate an explicit auxiliary energy function. By injecting task-specific first-order gradients during the reverse diffusion process, this guidance actively drives the optimization trajectory into deeper energy valleys. Together, these two stages ensure that the communication graph sampling process is rigorously guided by the energy landscape, guaranteeing the emergence of highly effective communication graphs that achieve the optimal trade-off between task utility and token cost.

This work bridges the gap between multi-agent coordination, generative modeling, and thermodynamic intuition. Our contributions are threefold: **Theoretical Framework:** We formalize multi-agent graph generation as an energy minimization process. This perspective shifts the paradigm from black-box search to continuous optimization, offering a principled explanation for the emergence of efficient collaboration as a natural consequence of energy reduction on a rugged energy landscape. **Methodology:** We pro-

pose **MAGE**, a novel score-based diffusion framework that navigates this landscape via learned gradient fields. By exploiting the mathematical duality between the score and the energy gradient, MAGE enables differentiable guidance for graph structures, ensuring stable convergence toward task-adaptive communication graphs. **Empirical Results:** Extensive experiments across six representative benchmarks demonstrate that MAGE achieves state-of-the-art performance. Furthermore, visualization analysis of the energy descent trajectories reveals that MAGE generates rational communication graphs mirroring the evolution of efficient organizations, validating our intuition.

2. PRELIMINARY

2.1. Problem Formulation

We consider a multi-agent system (MAS) composed of N agents, denoted by $V = \{v_1, \dots, v_N\}$. Given a task query $q \in \mathcal{Q}$ (e.g., a user prompt or problem specification), our objective is to synthesize an optimal, query-specific communication graph $G = (V, E)$, where the edge set E defines the information exchange channels among agents. The edge set E is generally represented by an adjacency matrix $A \in \{0, 1\}^{N \times N}$. To ensure executable and deterministic communication flows, we constrain G to be a directed acyclic graph (DAG) (Zhang et al., 2025b).

In thermodynamics, energy characterizes the instability of a system; analogously, in our context, $\mathcal{E}(G, q)$ quantifies the global inefficiency of a communication graph. It inherently unifies two conflicting objectives: the cost of coordination (e.g., token consumption) and the utility of task completion (e.g., pass rate). Under this energy-based formulation, the optimal structure corresponds to the minimum of the underlying energy landscape. Formally, the optimal graph for query q is defined as:

$$G^*(q) = \arg \min_G \mathcal{E}(G, q), \quad (1)$$

representing the state that aligns with the global minimum of the energy landscape envisioned in Section 1, thereby achieving an optimal trade-off between task utility and coordination cost.

2.2. Energy Minimization via Score-Based Diffusion

Directly solving Eq. (1) requires searching a combinatorial space, which is computationally intractable. To circumvent this, we relax the discrete optimization problem into a probabilistic sampling task by formulating communication graph generation as conditional energy-based modeling.

Conditional Energy-Based Objective. We bridge the optimization objective in Eq. (1) with generative modeling by defining a conditional probability distribution over commu-

nication graphs in the Boltzmann form:

$$p(G | q) = \frac{1}{Z(q)} \exp \left(-\frac{\mathcal{E}(G, q)}{\tau} \right), \quad (2)$$

where τ is a temperature parameter and $Z(q)$ is the intractable partition function. The rationale behind this formulation is that the exponential term assigns higher probability density to low-energy graphs. Thus, as $\tau \rightarrow 0$, the probability mass concentrates on the global energy minimum, making the task of finding the mode of the distribution mathematically equivalent to solving Eq. (1). However, directly sampling from this distribution is valid but intractable due to $Z(q)$. To circumvent this, we employ score-based generative modeling. Taking the gradient of the log-density:

$$\nabla_G \log p(G | q) = -\frac{1}{\tau} \nabla_G \mathcal{E}(G, q). \quad (3)$$

This derivation reveals a duality: the score function $\nabla_G \log p(G | q)$ is directly proportional to the negative gradient of the energy landscape. This equivalence is pivotal. It implies that the diffusion sampling process, driven by the score and typically annealed noise, functions mathematically as a form of stochastic gradient descent that minimizes the multi-agent system’s energy. Consequently, this formulation provides a mechanism to control the generation process: by leveraging the relationship between the score and the energy gradient, we can effectively steer the stochastic evolution of the graph toward the global energy minimum, thereby minimizing the energy $\mathcal{E}(G, q)$.

Conditional Score-Based Diffusion. Leveraging this duality, we employ a conditional score-based diffusion model s_θ to approximate the gradient field $\nabla_G \log p_t(G | q)$ across noise scales. The generation process is governed by the reverse-time Stochastic Differential Equation (SDE) conditioned on q :

$$dG = [f(G, t) - g(t)^2 \nabla_G \log p_t(G | q)] dt + g(t) d\bar{\mathbf{w}}, \quad (4)$$

where $f(G, t)$ and $g(t)$ denote the drift and diffusion coefficients of the forward process, respectively, and $\bar{\mathbf{w}}$ is a standard Wiener process in the reverse-time direction. By substituting the learned score function $s_\theta \approx \nabla_G \log p_t(G | q)$ into Eq. (4), the reverse diffusion process functions as a stochastic gradient descent over an implicit energy landscape captured by the data distribution, naturally steering the stochastic trajectory toward low-energy manifolds.

However, s_θ remains a statistical approximation of the data distribution. While effective at enforcing structural validity (directing the system away from high-energy, invalid regions), it captures the feasibility of a graph rather than its optimality. As a result, the score field lacks the fine-grained directional information required to enforce strictly

low-energy configurations, often converging to generic sub-optima that are valid but inefficient. Consequently, this implicit optimization serves primarily to position the system within the vicinity of low-energy graphs. This provides a valid initialization, laying the necessary groundwork for the subsequent injection of explicit energy guidance to drive the trajectory into the deeper energy valleys of the landscape.

3. METHODOLOGY: MAGE

Building upon the energy-based formulation established in the preliminaries, we propose **MAGE**, a score-based framework designed to generate task-adaptive communication graphs by navigating the underlying energy landscape. To implement this framework, we address two technical challenges: First, communication graphs are inherently hybrid, composed of discrete topological structures (adjacency matrices) and continuous semantic features (node embeddings). A critical challenge is designing a unified architecture capable of learning the joint score function over these heterogeneous modalities to capture their complex dependencies. Second, while the base diffusion model captures the general distribution of valid graphs, it does not inherently guarantee task-specific optimality. The challenge implies the need for a mechanism to actively steer the generation process toward global energy minima. To address these challenges, we employ a unified graph transformer to approximate the joint energy gradients and integrate an energy-guided sampling mechanism to synthesize low-energy, query-adaptive communication graphs, as illustrated in Figure 2.

3.1. Heterogeneous Joint Modeling

Joint Score Estimation. We represent the communication graph as $G = (\mathbf{A}, \mathbf{V})$, comprising discrete edge topologies $\mathbf{A}$ and continuous node embeddings $\mathbf{V}$. Rather than processing these modalities in isolation, we employ a Joint Graph Transformer to deeply aggregate the discrete topological structures with continuous semantic features. Based on this unified representation, we derive the joint score s_θ , which decomposes into two mutually conditioned components: **Node Score** $s_{\theta, \mathbf{V}}$: Derived from the aggregated node embeddings, representing the gradient field that guides the semantic evolution of agents. **Edge Score** $s_{\theta, \mathbf{A}}$: Computed via pairwise interactions of the aggregated features, representing the structural gradient that updates topological connections of communication graphs.

Architecture. We implement the joint score network (shown in Figure 2(a)) using a conditioned Graph Transformer (Yun et al., 2019). To capture the multi-faceted nature of the task, we first encode the query q into two distinct latent representations via a dual-path encoder: a *logic embedding* $\mathbf{z}_{\text{logic}}$ (encoding topological constraints) and a

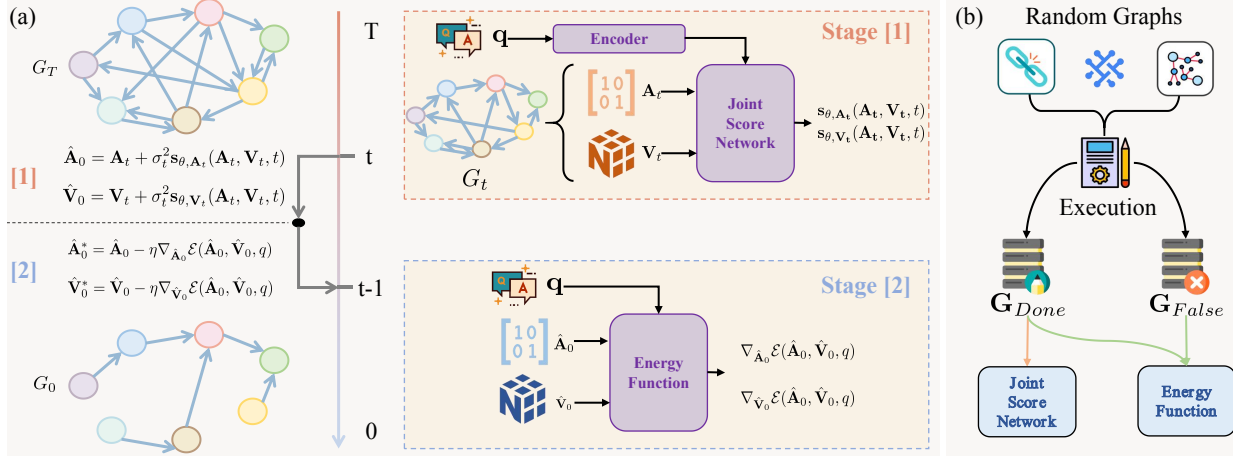


Figure 2. (a) Details of a single-step sampling process during energy guided sampling in Section 3.3. Stage [1] denotes the joint score estimation in Section 3.1, and Stage [2] denotes the energy function in Section 3.2. (b) The training data flow proceeds as follows: for each query q , we generate random graphs to facilitate MAS execution. Resulting graphs are then partitioned into successful examples (G_{Done}) and unsuccessful ones (G_{False}) based on task completion. Finally, we train the Joint Score Network and Energy Function separately using this collected data.

content embedding $\mathbf{z}_{\text{content}}$ (encoding node context). Additionally, the diffusion time step t is projected into a time embedding $\mathbf{t}_{\text{emb}}$ via a sinusoidal MLP. These conditions jointly modulate the Transformer to estimate the edge score and node score.

In our implementation, we modulate the message-passing topology by incorporating a structural bias $\mathbf{M}$ into the attention mechanism. The layer-wise update and final score prediction are formulated as:

$$\begin{aligned} \mathbf{M} &= \mathbf{A}_t + \text{Proj}_1(\mathbf{z}_{\text{logic}}), \\ \mathbf{H} &= \mathbf{V}_t + \text{Attention}(\text{LN}(\mathbf{V}_t), \text{bias} = \mathbf{M}), \\ \mathbf{V}_{\text{out}} &= \mathbf{H} + \text{FFN}(\text{LN}(\mathbf{H}) + \text{Proj}_2(\mathbf{z}_{\text{content}}) + \mathbf{t}_{\text{emb}}), \end{aligned} \quad (5)$$

where LN denotes LayerNorm, and $\text{Proj}_{1,2}$ are projection MLPs. The structural bias $\mathbf{M}$ guides the attention based on the noisy adjacency $\mathbf{A}_t$ and logic constraints, while the time embedding $\mathbf{t}_{\text{emb}}$ and content embedding $\mathbf{z}_{\text{content}}$ inject temporal and semantic context into the feature transformation. Subsequently, the scores are derived from the refined features $\mathbf{V}_{\text{out}}$ via:

$$\begin{aligned} \mathbf{s}_{\theta, \mathbf{V}_t} &= \text{MLP}_{\text{node}}(\mathbf{V}_{\text{out}}), \\ [\mathbf{s}_{\theta, \mathbf{A}_t}]_{ij} &= \text{Head}_{\text{edge}}(\mathbf{v}_{\text{out}, i}, \mathbf{v}_{\text{out}, j}), \end{aligned} \quad (6)$$

where $\text{Head}_{\text{edge}}$ is a pairwise interaction layer. This architecture ensures that the edge score $\mathbf{s}_{\theta, \mathbf{A}}$ is strictly conditioned on the globally contextualized node states, maintaining consistency between topology and nodes.

Training. We train the joint score network to approximate the joint score function of the data distribution (shown in

Figure 2(b)). The network is optimized to simultaneously recover the discrete topological structure and continuous node attributes from their noisy counterparts. Following (Song et al., 2021), we formulate the training objective as minimizing the weighted Mean Squared Error (MSE):

$$\begin{aligned} \mathcal{L}_{\text{train}} &= \mathbb{E}_{t, G_0, G_t} \left[\lambda_A \|\mathbf{s}_{\theta, \mathbf{A}_t} - \nabla_{\mathbf{A}_t} \log p(G_t | G_0)\|_F^2 \right. \\ &\quad \left. + \lambda_V \|\mathbf{s}_{\theta, \mathbf{V}_t} - \nabla_{\mathbf{V}_t} \log p(G_t | G_0)\|_F^2 \right], \end{aligned} \quad (7)$$

where G_0 denotes the ground-truth communication graph, G_t represents the perturbed state at time t , and λ_A, λ_V are weighting coefficients balancing the two modalities. Also, here we adopt classifier-free guidance to ensure the query q is correctly modeled as the condition (Ho & Salimans, 2022).

3.2. Task-Dependent Energy Function

While the learned score model $\mathbf{s}_{\theta}$ ensures that generated graphs lie on the manifold of valid communication structures, it inherently approximates an indiscriminate empirical distribution. Since the training set encompasses all successful executions regardless of their efficiency, the learned density assigns high probability to a vast space of feasible but redundant executions. Consequently, the score network perceives a relatively flat energy landscape, lacking the discriminative gradients to distinguish between local optimum (e.g., graphs generating unnecessary tokens) and the true global optimum.

To navigate this landscape effectively, we explicitly characterize the latent energy potential governing the system. This formulation balances two competing forces: minimizing organizational friction (communication overhead) and max-

imizing task utility (problem-solving accuracy). Formally:

$$\mathcal{E}(G, q) = \lambda_{\text{cost}} \mathcal{C}(G) - \mathcal{U}(G, q), \quad (8)$$

where $\mathcal{C}(G)$ represents the *collaboration cost* (e.g., total token consumption or edge density), $\mathcal{U}(G, q)$ denotes the *task utility* (i.e., the quality of the response to query q), and λ_{cost} is a hyperparameter governing the trade-off. Under this formulation, minimizing energy corresponds to identifying the graph that achieves the highest utility with the least resource consumption.

Architecture. We implement the energy model, parameterized by ϕ , using a Graph Attention Network (Yun et al., 2019; Jiang et al., 2025). This network serves as a differentiable critic, taking the adjacency matrix $\mathbf{A}$, node features $\mathbf{V}$, and task query q as inputs. To enable dynamic trade-offs during inference, the model is designed with dual output heads that independently regress the expected collaboration cost and task utility: $\hat{c}, \hat{u} = \mathcal{E}_\phi(\mathbf{A}, \mathbf{V}, q)$.

Training. The energy model is trained to approximate the ground-truth performance landscape via supervised learning. As illustrated in Figure 2(b), we construct a training dataset $\mathcal{D}_{\text{energy}}$ comprising diverse communication graphs along with their ground-truth labels (i.e., empirical execution cost and success status). The model parameters ϕ are optimized to minimize the Mean Squared Error (MSE) between the predicted components and the ground-truth values. Once trained, $\mathcal{E}_\phi$ acts as a surrogate function, providing differentiable gradients $\nabla_{G_t} \mathcal{E}(G_t, q)$ to guide the diffusion sampling process toward low-energy regions characterized by high utility and low redundancy.

3.3. Energy-Guided Sampling

To steer the generative trajectory toward the global optimum, we augment the diffusion sampling with gradients derived from the learned energy model $\mathcal{E}_\phi$. At each diffusion timestep t , we first estimate the clean graph $\hat{G}_0 = (\hat{\mathbf{A}}_0, \hat{\mathbf{V}}_0)$ from the current noisy state. This reconstruction is computed by applying Tweedie’s formula to the score prediction s_θ , as illustrated in Stage [1] of Figure 2(a).

Crucially, to propagate energy gradients back through the discrete structural component, we apply the Gumbel-Sigmoid relaxation (Jang et al., 2017) to the predicted adjacency logits $\hat{\mathbf{A}}_0$. This transformation yields a continuous approximation $\tilde{\mathbf{A}}_0 \in [0, 1]^{N \times N}$, enabling end-to-end differentiability. We then evaluate the relaxed structure $\tilde{\mathbf{A}}_0$ and node features $\hat{\mathbf{V}}_0$ via the energy function to compute the gradients via the chain rule:

$$\begin{aligned} \nabla_{\hat{\mathbf{A}}_0} \mathcal{E} &= \nabla_{\tilde{\mathbf{A}}_0} \mathcal{E}_\phi(\tilde{\mathbf{A}}_0, \hat{\mathbf{V}}_0, q) \cdot \frac{\partial \tilde{\mathbf{A}}_0}{\partial \hat{\mathbf{A}}_0}, \\ \nabla_{\hat{\mathbf{V}}_0} \mathcal{E} &= \nabla_{\hat{\mathbf{V}}_0} \mathcal{E}_\phi(\tilde{\mathbf{A}}_0, \hat{\mathbf{V}}_0, q). \end{aligned} \quad (9)$$

Leveraging these gradients, we perform a single step of gradient descent to refine the estimated clean graph toward lower-energy configurations (i.e., Stage [2] of Figure 2(a)):

$$\hat{\mathbf{A}}_0^* = \hat{\mathbf{A}}_0 - \eta \nabla_{\hat{\mathbf{A}}_0} \mathcal{E}, \quad \hat{\mathbf{V}}_0^* = \hat{\mathbf{V}}_0 - \eta \nabla_{\hat{\mathbf{V}}_0} \mathcal{E}, \quad (10)$$

where η is the guidance step size. Finally, the refined clean estimate $\hat{G}_0^* = (\hat{\mathbf{A}}_0^*, \hat{\mathbf{V}}_0^*)$ is substituted into the diffusion posterior to sample the previous state G_{t-1} . This mechanism effectively enables the energy model to actively guide both topological and node features, overcoming the constraints of discreteness. We provide a detailed theoretical analysis of energy-guided sampling in Appendix C.

4. EXPERIMENTS

4.1. Experiment Setup

Benchmarks. To evaluate the MAGE framework, we prioritize reasoning-intensive domains over general assistant tasks. This selection is strategic: MAGE is designed to optimize the complex collaborative structures and logical dependencies. Consequently, we exclude datasets primarily characterized by heavy tool utilization or environment interaction, where the performance bottleneck stems from execution capabilities rather than the efficacy of the communication graph. Hence, we conduct experiments on six diverse benchmarks covering mathematical reasoning and code generation: GSM8K (Cobbe et al., 2021), MATH (Hendrycks et al.), MultiArith (Roy & Roth, 2015), SVAMP (Patel et al., 2021), HumanEval (Chen et al., 2021), and MBPP (Austin et al., 2021). Following prior work (Zhang et al., 2025a), we sample 50 instances from each benchmark to construct the training set, with the remaining instances used for evaluation. Dataset statistics and the detailed dataset construction procedure can be found in Appendix A.

Baselines. We benchmark MAGE against a comprehensive set of methods spanning three groups: (1) single-agent baselines, including CoT (Wei et al., 2022) and SC-CoT (Wang et al., 2023); (2) human-designed multi-agent frameworks like LLM-Debate (Du et al., 2024), LLM-Blender (Jiang et al., 2023), and DyLAN (Liu et al., 2024c); and (3) automated multi-agent frameworks, including GPTSwarm (Zhuge et al., 2024), ADAS (Hu et al., 2025), AFLOW (Zhang et al., 2025d), G-Designer (Zhang et al., 2025c), ARG-Designer (Li et al., 2026), MaAS (Zhang et al., 2025a), and GTD (Jiang et al., 2025).

4.2. Overall Performance

We compare MAGE with 3 groups of current baselines on commonly used benchmarks. The result is illustrated in Table 1. From the experimental results, MAGE achieves state-of-the-art performance across all tasks, and we can summarize the following findings:

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Method	GSM8K	MATH	MultiArith	SVAMP	HumanEval	MBPP	Avg.
Vanilla	87.45	60.20	96.85	86.67	87.08	71.83	81.68
CoT	87.10 $\downarrow 0.35$	62.67 $\uparrow 2.47$	96.31 $\downarrow 0.54$	87.40 $\uparrow 0.73$	88.13 $\uparrow 1.05$	72.00 $\uparrow 0.17$	82.27 $\uparrow 0.59$
SC-CoT	87.57 $\uparrow 0.12$	62.22 $\uparrow 2.02$	96.58 $\downarrow 0.27$	88.12 $\uparrow 1.45$	88.60 $\uparrow 1.52$	73.60 $\uparrow 1.77$	82.78 $\uparrow 1.10$
LLM-Debate	89.47 $\uparrow 2.02$	61.11 $\uparrow 0.91$	97.33 $\uparrow 0.48$	89.04 $\uparrow 2.37$	88.68 $\uparrow 1.60$	70.29 $\downarrow 1.54$	82.65 $\uparrow 0.97$
LLM-Blender	88.35 $\uparrow 0.90$	65.55 $\uparrow 5.35$	97.29 $\uparrow 0.44$	87.33 $\uparrow 0.66$	88.80 $\uparrow 1.72$	77.05 $\uparrow 5.22$	84.06 $\uparrow 2.38$
DyLAN	89.98 $\uparrow 2.53$	64.20 $\uparrow 4.00$	97.12 $\uparrow 0.27$	88.67 $\uparrow 2.00$	90.42 $\uparrow 3.34$	77.30 $\uparrow 5.47$	84.62 $\uparrow 2.94$
GPTSwarm	89.14 $\uparrow 1.69$	61.55 $\uparrow 1.35$	96.79 $\downarrow 0.06$	88.67 $\uparrow 2.00$	89.32 $\uparrow 2.24$	77.43 $\uparrow 5.60$	83.82 $\uparrow 2.14$
ADAS	86.12 $\downarrow 1.33$	58.22 $\downarrow 1.98$	96.02 $\downarrow 0.83$	88.43 $\uparrow 1.76$	84.19 $\downarrow 2.89$	68.13 $\downarrow 3.70$	80.19 $\downarrow 1.49$
AFLOW	91.16 $\uparrow 3.71$	64.20 $\uparrow 4.00$	96.22 $\downarrow 0.63$	88.33 $\uparrow 1.66$	90.93 $\uparrow 3.85$	81.67 $\uparrow 9.84$	85.42 $\uparrow 3.74$
MaAS	92.30 $\uparrow 4.85$	65.80 $\uparrow 5.60$	98.80 $\uparrow 1.95$	89.67 $\uparrow 3.00$	92.85 $\uparrow 5.77$	82.17 $\uparrow 10.34$	86.93 $\uparrow 5.25$
ARG-Designer	93.90 $\uparrow 6.45$	64.00 $\uparrow 3.80$	98.93 $\uparrow 2.08$	94.50 $\uparrow 7.83$	91.74 $\uparrow 4.66$	82.80 $\uparrow 10.97$	87.65 $\uparrow 5.97$
GTD	94.14 $\uparrow 6.69$	68.44 $\uparrow 8.24$	98.88 $\uparrow 2.03$	91.33 $\uparrow 4.66$	91.46 $\uparrow 4.38$	82.50 $\uparrow 10.67$	87.79 $\uparrow 6.11$
MAGE (Ours)	94.35 $\uparrow 6.90$	72.22 $\uparrow 12.02$	99.22 $\uparrow 2.37$	94.81 $\uparrow 8.14$	94.23 $\uparrow 7.15$	84.57 $\uparrow 12.74$	89.90 $\uparrow 8.22$

Table 1. Performance comparison on various benchmarks. The main numbers represent accuracy (%), and the subscripts indicate the performance gap relative to the **Vanilla** baseline.

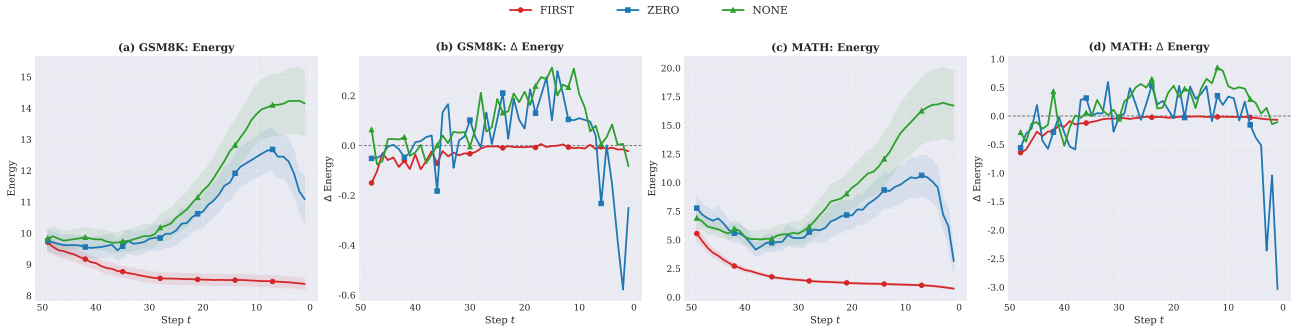


Figure 3. **Analysis of Energy Dynamics and Convergence.** We compare the training dynamics of FIRST (Our method), ZERO (Jiang et al., 2025), and NONE on GSM8K and MATH datasets. (a, c) illustrate the energy (E) across diffusion steps t , where FIRST achieves the lowest final energy. (b, d) display the step-wise energy reduction (Δ Energy). The dashed line at $y = 0$ marks the optimization boundary; FIRST exhibits a significantly sharper initial descent and sustained effective optimization compared to baselines.

(1) MAGE significantly outperforms human-designed multi-agent frameworks by a margin of at least 6%. Similarly, existing automated multi-agent frameworks (e.g., MaAS) also generally outperform these baselines. In all, this substantiates the conclusion that dynamically altering agent communication graphs tailored to specific queries is beneficial. (2) MAGE surpasses existing frameworks like ARG-Designer, GPTSwarm, and MaAS. MAGE utilizes a holistic graph diffusion mechanism derived from thermodynamic intuition. By directly accessing the energy landscape’s gradients rather than merely observing energy values, MAGE achieves a more efficient optimization. (3) Compared to LLM-generated workflow frameworks (e.g., AFLOW and ADAS), our approach eliminates the need for communication graph reflection, thereby reducing the cost of the graph generation process while still surpassing these baselines.

4.3. Guidance Analysis

To empirically validate the efficacy of the Energy-Guided Sampling, we compare our approach against two representative baselines alongside our proposed method, FIRST, which leverages first-order gradients to actively steer the generation trajectory. The first baseline is ZERO, a zeroth-order strategy (Jiang et al., 2025) that updates G_{t-1} using the lowest-energy candidate from random perturbations; crucially, this baseline serves as a rigorous proxy for prevalent black-box optimization paradigms (e.g., evolutionary search). The second baseline is NONE, which relies exclusively on query conditioning, devoid of any runtime energy guidance. Experiments on the GSM8K and MATH datasets track the energy trajectories of 50 generated samples at each diffusion step based on the pre-trained energy function.

As illustrated in Figure 3, we plot both the energy land-

Variant	GSM8K	MBPP	MATH
MAGE (Ours)	94.35	84.57	72.22
w/o Joint	92.27 $\downarrow 2.08$	83.75 $\downarrow 0.82$	70.09 $\downarrow 2.13$
w/o Guidance	93.85 $\downarrow 0.50$	83.78 $\downarrow 0.79$	71.43 $\downarrow 0.79$

Table 2. The result of the ablations including w/o Joint and w/o Guidance. The values in green parentheses indicate the performance drop compared to the full MAGE model.

scape and the step-wise energy reduction (Δ). The results demonstrate that FIRST achieves the smoothest convergence trajectory and the lowest final energy. This stability indicates that explicit energy guidance effectively steers the optimization of the communication graph, avoiding the erratic updates typical of stochastic approximations. In contrast, ZERO exhibits significantly higher variance across samples. Since zeroth-order methods rely on stochastic estimation via random perturbations, the guidance direction is inherently imprecise. This leads to volatile Δ fluctuations, where the process oscillates between effective reductions and detrimental energy increases. Finally, NONE lacks explicit guidance; while it initially reduces energy based on the query prior, it eventually degrades (energy rises) in later steps due to a tendency to regress toward the mean. Taken together, these observations, along with our theoretical analysis in Appendix C, demonstrate the effectiveness of MAGE in synthesizing low-energy communication graphs.

4.4. Further Analysis

Ablation Study. We present the ablation study in Table 2. As observed, removing the joint training mechanism (*w/o Joint*) results in a clear performance decline across all datasets (e.g., a **2.13%** drop on MATH). Similarly, removing the guidance mechanism (*w/o Guidance*) also leads to consistent accuracy losses (e.g., **0.79%** on MBPP and MATH). The full MAGE model consistently outperforms both variants, demonstrating that both the joint training strategy and the energy-guided guidance are essential.

Cost Analysis. The total cost is a critical metric in MAS, as it governs not only financial expenditure but also the quality of information synthesis. Excessive communication cost can lead to the dilution of critical information within an overextended context (Gao et al., 2025), thereby degrading the overall effectiveness of the communication graph. To assess this effect, we compare the token consumption of MAGE with several state-of-the-art methods on the GSM8K benchmark (see Table 3). All evaluated methods exhibit moderate token consumption, which can be largely attributed to their mechanisms for regulating communication graph sparsity. Notably, MAGE achieves the lowest token and financial cost while simultaneously attain-

Method	Prmpt. (10^6)	Compl. (10^6)	Cost (\$)	ACC (%)
GPTSwarm	7.29	2.11	2.36	89.14
MaAS	3.51	2.28	1.90	92.30
ARG-Designer	2.75	1.82	1.51	93.90
GTD	2.62	1.81	1.48	94.14
MAGE	2.59	1.73	1.43	94.35

Table 3. Comparison of computational cost and performance. "Prmpt." and "Compl." denote the total number of prompt and completion tokens, respectively.

Method	10	20	30	40	50
MAGE-First	72.08	74.33	71.43	71.88	72.22
MAGE-Zero	70.54	69.19	70.31	71.42	70.31

Table 4. The effect of diffusion steps

ing the highest accuracy. This efficiency arises from our refined energy-guided sampling framework, which effectively prunes redundant communication and enhances the information density of multi-agent collaboration.

Diffusion Step Analysis. Table 4 investigates the impact of different diffusion steps on performance. It is evident that MAGE-First consistently outperforms MAGE-Zero across all step settings. Notably, at 20 steps, MAGE-First achieves a peak performance of 74.33, significantly surpassing MAGE-Zero (69.19). Corroborating with the previously discussed energy descent curve, these results highlight the superior stability of the first-order approach. While the zeroth-order method tends to struggle with directional uncertainty on the energy landscape, the incorporation of first-order gradient information in MAGE-First acts as an effective guide. This property ensures a more distinct and robust convergence towards lower energy states, thereby preventing the instability and performance fluctuations observed in the zeroth-order baseline.

Guidance Scale η Analysis. We investigate the sensitivity of our method to the guidance scale η by varying it from 1.0 to 5.0 across three datasets, as shown in Figure 4. The results reveal no significant linear correlation between the magnitude of η and the final performance. For instance, on MBPP, the accuracy fluctuates marginally between 84.31% and 87.23% without a drastic drop-off even at higher scales. Similarly, HumanEval maintains high performance across the spectrum. This observation underscores the effectiveness of our energy-guided mechanism. The insensitivity to the scale hyperparameter suggests that the gradient direction derived from the energy function is well-aligned with the target distribution. Because the update direction provides a reliable guidance for the generation trajectory, the method

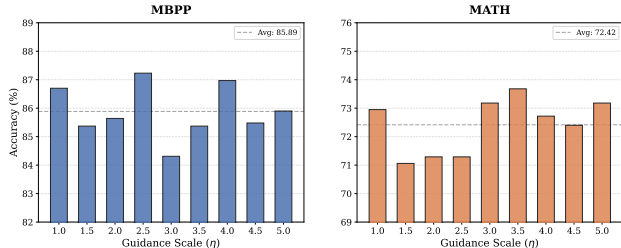


Figure 4. Impact of the guidance scale η . The consistent performance across a wide range ($\eta \in [1, 5]$) demonstrates the stability of our energy-guided sampling.

becomes less dependent on the precise tuning of the step magnitude, thereby demonstrating superior stability and ease of deployment compared to methods that require delicate scale calibration.

5. RELATED WORK

LLM-Based Multi-Agent Systems. The rapid advancement of Large Language Models (LLMs) has fundamentally reshaped artificial intelligence (Achiam et al., 2023), driving a paradigm shift from static response generation toward autonomous agents capable of sophisticated reasoning (Yao et al., 2022; Guo et al., 2025; Liu et al., 2025) and effective tool use (Ma et al., 2025; Qin et al., 2024). While monolithic agents demonstrate strong performance across a broad range of tasks (Madaan et al., 2023; Wang et al., 2024; Park et al., 2023; Liu et al., 2024b), the field is increasingly pivoting toward Collaborative Multi-Agent Systems (MAS) (Zhuge et al., 2024; Zhang et al., 2025c;b) to transcend individual capability boundaries and tackle complex, long-horizon problems (Li et al., 2024; Qian et al., 2025). In such systems, the communication graph serves as a critical determinant of task utility, making the design of interaction structures a key research priority (Zhang et al., 2025d; Zhuge et al., 2024). However, early literature has predominantly relied on static, manually crafted topologies, ranging from layered architectures (Wang et al., 2025a) and linear sequential workflows (Hong et al., 2023) to tree-based reasoning structures (Du et al., 2024; Jiang et al., 2023) and hierarchical administrative setups (Wu et al., 2024). Despite their foundational contributions, the inherent rigidity of these fixed structures limits their adaptability across diverse tasks, frequently resulting in sub-optimal performance.

Automatic Communication Graph Optimization. Recent research has pivoted towards learning adaptive communication graphs. Initial approaches focused on pruning dense graphs, such as AgentPrune (Zhang et al., 2025b), or optimizing edge probabilities via reinforcement learning, as seen in GPTSwarm (Zhuge et al., 2024) and MaAS (Zhang et al., 2025a). To transcend these adjustments, advanced methods have adopted generative paradigms. For instance,

G-Designer (Zhang et al., 2025c) employs GNN-based autoencoders, while ARG-Designer (Li et al., 2026) utilizes autoregressive networks (Hochreiter & Schmidhuber, 1997) for sequential graph construction. Similarly, methods like ADAS (Hu et al., 2025) and AFLOW (Zhang et al., 2025d) harness the capabilities of LLMs to directly optimize communication graphs. More recently, GTD (Jiang et al., 2025) offers superior capability in modeling complex edge distributions via diffusion models. However, a critical gap remains. While existing generative approaches excel at constructing valid topologies, they operate primarily under a zeroth-order paradigm, relying on scalar feedback (e.g., rewards) or data imitation rather than explicit gradient guidance. Consequently, they fundamentally neglect the underlying energy landscape of multi-agent collaboration, thereby failing to guide the generation of lower-energy communication graphs.

Score-Based Graph Diffusion Models. Recently, foundational works like DiGress (Vignac et al., 2023) and GDSS (Jo et al., 2022) have established the capability of SDE-based frameworks for capturing complex graph dependencies. While recent advancements have introduced sophisticated guidance mechanisms (Liu et al., 2024a; Wang et al., 2025b; Kerby & Moon, 2024), these methods are predominantly tailored for molecular discovery, where the objective is to satisfy static structural constraints (e.g., valency rules and chemical stability). They cannot be directly transplanted to MAS, as optimizing agent communication graphs involves navigating a dynamic, task-specific energy landscape. Unlike molecular generation, which focuses on structural validity, MAS graph design requires minimizing collaboration cost and maximizing task utility, necessitating a shift from pattern matching to energy descent.

6. CONCLUSION

In this paper, we presented a novel framework for optimizing communication graphs in multi-agent systems by bridging graph generation with the energy landscape. We formulated the construction of communication graphs as an energy minimization process, establishing a theoretical foundation for the task. Specifically, we employed a score-based diffusion model to act as the communication graph generator. Under the guidance of a theoretically grounded energy function, our model iteratively refines random, high-energy initializations into task-specific, low-energy, and highly efficient communication graphs. Distinct from existing methods, our energy function offers superior theoretical guarantees regarding guidance effectiveness. Both our theoretical analysis and extensive empirical results demonstrate the validity and efficacy of this energy landscape formulation in achieving efficient multi-agent coordination.

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Impact Statement

This work introduces MAGE, a framework that leverages thermodynamic intuition to optimize the structural efficiency of Multi-Agent Systems (MAS). By formalizing coordination as an energy minimization process, our approach significantly mitigates the redundancy and computational overhead (e.g., token consumption) inherent in large-scale collaboration, thereby contributing to more sustainable and scalable AI deployments. This capability to generate streamlined communication graphs holds significant potential for advancing complex domains. While we acknowledge the necessity for the responsible deployment of advanced coordination technologies, our primary contribution lies in addressing the fundamental bottlenecks that currently hinder the emergence of collective intelligence in LLM-based multi-agent systems.

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A. DATA STATISTICS

In this paper, we evaluate the proposed MAGE framework and compare it with state-of-the-art baselines on several widely used public benchmarks. To train the Energy Function and the diffusion model, we sample 50 instances from each dataset. Following the methodology in (Jiang et al., 2025; Zhang et al., 2025c), we generate random communication graphs—such as chain, tree, and star structures—to execute these 50 queries. Specifically, we generate 6 graphs per sample, resulting in a total of 300 graphs. We utilize the entire set of 300 graphs (including both successful and failed executions) to train the energy function, while only the successful instances were used to train the diffusion denoising network. Finally, during the testing phase, we employ energy-guided sampling with the trained models to generate optimal communication graphs for the remaining test samples.

Table 5. Summary of benchmarks, task types, metrics, and data splits.

Benchmark	Task	Metric	Training	Testing
GSM8K	Math	Accuracy	50	1269
HumanEval	Code	Pass@1	50	110
MATH	Math	Accuracy	50	450
MBPP	Code	Pass@1	50	377
MultiArith	Math	Accuracy	50	130
SVAMP	Math	Accuracy	50	950

B. IMPLEMENTATION DETAILS.

In this paper, we adopt GPT-4o-mini-0718 as the backbone model for all agents. The communication graph generator is implemented using a Graph Transformer-based diffusion model. Each node feature (i.e., the feature of a single node in $\mathbf{V}$) is represented by a 384-dimensional embedding, while task conditions are encoded into 384-dimensional embeddings derived from role descriptions and task queries using all-MiniLM-L6-v2. These embeddings are projected into a 128-dimensional hidden state, and diffusion timesteps are encoded with a 128-dimensional time embedding. The diffusion process is conducted for 50 steps, and the model is trained for 10 epochs on up to 50 generated graph instances. During sampling, energy-function guidance introduced in Stage [2] is applied, with the scale factor η set to 2.0, to steer the generation toward low-energy communication graphs. The core code is available in <https://github.com/ghzha0/mage>.

C. THEORETICAL ANALYSIS

In this section, we provide a theoretical justification for our proposed method, **MAGE**. Although our data G consists of a graph structure with discrete edges E and continuous node features V , these specific structural details do not alter the underlying derivation. We assume that discrete structures are represented via continuous relaxations, as is standard in diffusion-based graph generation. Thus, for the purpose of analysis, we abstract the target as a state vector x in a high-dimensional continuous space. We demonstrate that back-propagating the energy gradient through the score network is mathematically equivalent to projecting the guidance signal onto the data manifold via the Jacobian matrix.

Let x_t denote the noisy state at time step t , and let $s_\theta(x_t, q)$ be the pre-trained score network that approximates the score of the data distribution, i.e., $s_\theta(x_t, q) \approx \nabla_{x_t} \log p_t(x_t|q)$. The clean data estimate $\hat{x}_0(x_t)$ is derived from the score via Tweedie’s formula:

$$\hat{x}_0(x_t) = \frac{x_t + \sigma_t^2 s_\theta(x_t, q)}{\alpha_t}, \quad (11)$$

where α_t, σ_t are coefficients determined by the noise schedule. The energy function $\mathcal{E}(\cdot, q)$ is defined on this estimated clean data domain. Our objective is to sample from the guided conditional distribution:

$$\tilde{p}_t(x_t|q) \propto p_t(x_t|q) \exp(-\eta \mathcal{E}(\hat{x}_0(x_t), q)), \quad (12)$$

where η is the guidance scale.

C.1. Manifold-Aware Guidance via Jacobian

To sample from $\tilde{p}_t(x_t|q)$ using diffusion models, we need to derive its corresponding score function. The following theorem establishes the optimal form of the drift correction.

Theorem C.1 (Form of Energy-based Guidance). *For a diffusion process governed by the reverse-time SDE, the optimal score correction required to sample from the energy-guided distribution $\tilde{p}_t(x_t|q)$ is given by the gradient of the energy composed with the reconstruction mapping. Specifically, the modified score function $\tilde{s}_\theta(x_t, q)$ satisfies:*

$$\tilde{s}_\theta(x_t, q) = s_\theta(x_t, q) - \eta \cdot \nabla_{x_t} \mathcal{E}(\hat{x}_0(x_t), q). \quad (13)$$

Expanding the gradient term via the chain rule reveals the dependence on the score network’s Jacobian:

$$\nabla_{x_t} \mathcal{E}(\hat{x}_0(x_t), q) = \mathbf{J}_{\hat{x}_0}(x_t)^\top \nabla_{\hat{x}_0} \mathcal{E}(\hat{x}_0, q), \quad (14)$$

where $\mathbf{J}_{\hat{x}_0}(x_t) = \frac{\partial \hat{x}_0(x_t)}{\partial x_t}$ is the Jacobian of the reconstruction function.

Proof. The target score function is defined as the gradient of the log-density of the guided distribution $\tilde{p}_t$. By linearity of the gradient operator:

$$\begin{aligned} \nabla_{x_t} \log \tilde{p}_t(x_t|q) &= \nabla_{x_t} [\log p_t(x_t|q) - \eta \mathcal{E}(\hat{x}_0(x_t), q) - \log Z_q] \\ &\approx s_\theta(x_t, q) - \eta \nabla_{x_t} (\mathcal{E}(\hat{x}_0(x_t), q)). \end{aligned} \quad (15)$$

Applying the chain rule to the composite function $\mathcal{E} \circ \hat{x}_0$:

$$\nabla_{x_t} \mathcal{E}(\hat{x}_0(x_t), q) = \left(\frac{\partial \hat{x}_0(x_t)}{\partial x_t} \right)^\top \nabla_{\hat{x}_0} \mathcal{E}(\hat{x}_0, q). \quad (16)$$

From Tweedie’s formula (Eq. 11), the Jacobian of the reconstruction is $\mathbf{J}_{\hat{x}_0}(x_t) = \frac{1}{\alpha_t} (I + \sigma_t^2 \mathbf{J}_{s_\theta}(x_t))$, where $\mathbf{J}_{s_\theta}(x_t)$ is the Jacobian of the score network. Substituting this back yields the final expression. $\square$

The presence of the Jacobian matrix in the gradient term carries a profound geometric interpretation.

Remark C.2 (Manifold-Consistent Projection). The term $\mathbf{J}_{\hat{x}_0}(x_t)^\top \nabla_{\hat{x}_0} \mathcal{E}(\hat{x}_0, q)$ acts as a geometric projection. Since the reconstruction $\hat{x}_0(x_t)$ maps the noisy ambient space to the data manifold $\mathcal{M}$, its Jacobian matrix captures the local tangent space of the manifold. Specifically, because $\hat{x}_0(x_t)$ relies on the score network s_θ , the back-propagated gradient $\nabla_{x_t} \mathcal{E}$ is implicitly filtered by $\mathbf{J}_{s_\theta}^\top$, biasing the guidance signal toward directions corresponding to valid structural variations modeled by the pre-trained diffusion model, effectively suppressing off-manifold noise.

Remark C.3 (Inefficiency of Zeroth-order Guidance). In contrast, zeroth-order methods (e.g., selection-based guidance (Jiang et al., 2025)) rely on estimating the guidance direction via random exploration without access to the gradient information from s_θ . In high-dimensional spaces, random perturbations have vanishing expected alignment with the true descent direction, leading to extremely inefficient gradient estimation (Flaxman et al., 2005). This phenomenon, known as the curse of dimensionality, renders zeroth-order methods computationally intractable for guiding complex graph generation, as they fail to leverage the manifold geometry encoded in $\mathbf{J}_{s_\theta}$.

Collectively, these theoretical insights, along with the experimental results in Section 4.2 and Section 4.3, comprehensively validate the effectiveness and robustness of our proposed theoretical framework and MAGE.

D. EXTRA EXPERIMENTS

To evaluate the framework’s adaptability to the rapidly evolving landscape of Large Language Models (LLMs), we conducted a “cold-start” evaluation on Qwen3-32B and LLaMA-4-Scout. Notably, these models were released subsequent to GPT-4o-mini and were not included during the training phase. As illustrated in Table 6, MAGE demonstrates robust generalization capabilities. Despite the lack of specific fine-tuning, the framework successfully boosts performance across most scenarios, particularly on the GSM8K benchmark. This validates the practical necessity for multi-agent communication graphs to remain effective when facing frequent model updates. However, we observe a performance degradation for LLaMA-4 on the MBPP dataset (from 83.13% to 77.93%). We attribute this anomaly to *context window limitations*. The extensive interaction history generated by the multi-agent framework may exceed the effective context capacity of LLaMA-4 in coding tasks, leading to information loss or confusion during inference.

Table 6. Extra Performance comparison across different backbones on GSM8K and MBPP datasets.

Dataset	Method	Qwen3-32B	LlaMA-4-Scout	GPT-4o-mini
GSM8K	Vanilla	82.49	94.92	87.45
	MAGE	95.89	96.04	94.35
MBPP	Vanilla	83.84	83.13	71.83
	MAGE	85.90	77.93	84.57

As shown in Table 7, MAGE benefits from a larger agent pool within a moderate range: Performance improves from 2 to 10 agents, but drops slightly at 15 while token cost increases, suggesting reasonable scalability at moderate scales but diminishing returns beyond that range.

Table 7. MAGE performance with varying numbers of agents.

Agent Nums	2	5	10	15
MBPP Performance	0.8430	0.8644	0.8697	0.8537
Prompt Tokens	310,049	576,769	1,130,546	1,830,913
Completion Tokens	93,091	160,070	193,027	230,442

The parameter λ directly controls the trade-off between cost and utility. Table 8 shows that neither overly small nor overly large values of λ are desirable, and a moderate value is needed to achieve a balance between efficiency and performance.

Table 8. Sensitivity analysis of the cost-control parameter on HumanEval.

Parameter (λ)	Prompt Tokens	Completion Tokens	Performance
0.001	220,127	75,600	0.846
0.005	219,568	76,985	0.875
0.01	216,754	72,831	0.933
0.05	203,903	68,309	0.904

E. Prompts

Prompt of Mathematical Roles

MathSolver You are a master of mathematics. Your task is to solve the given math word problem and provide a step-by-step solution. Focus on accuracy and clear explanation. You must output the final numerical answer in the format '####Answer: <number>'.

ProblemDecomposer You are an expert at breaking down complex problems. Your role is to analyze the math word problem, identify the key variables, and outline the sequence of steps required to solve it. Do not solve the problem yourself, only provide the plan.

CalculationChecker You are a meticulous calculation checker. You will be given a proposed solution to a math problem. Your job is to verify every calculation, step by step, and state whether the final answer is correct or not. Point out any specific errors you find.

ProgrammingExpert You are a programming expert who solves math problems by writing simple Python code. You will be given a math word problem. Write a Python script that calculates the answer, and then execute it to get the final result. The final line of your response should be the numerical answer.

FinalRefer You are the final arbiter. Your task is to review the proposed solutions and select the most accurate and well-reasoned answer.

Prompt of Coding Roles

AlgorithmDesigner You are an algorithm designer. You will be given a function signature and its docstring by the user. You need to specify the specific design of the algorithm, including the classes that may be defined and the functions used. You need to generate the detailed documentation, including explanations of the algorithm, usage instructions, and API references. When the implementation logic is complex, you can give the pseudocode logic of the main algorithm. I hope your reply will be more concise. Preferably within fifty words. Don't list too many points.

ProjectManager You are a project manager. You will be given a function signature and its docstring by the user. You are responsible for overseeing the overall structure of the code, ensuring that the code is structured to complete the task. Implement code concisely and correctly without pursuing over-engineering. You need to suggest optimal design patterns to ensure that the code follows best practices for maintainability and flexibility. You can specify the overall design of the code, including the classes that need to be defined (maybe none) and the functions used (maybe only one function). I hope your reply will be more concise. Preferably within fifty words. Don't list too many points.

ProgrammingExpert You are a programming expert. You will be given a function signature and its docstring by the user. Carefully read the docstring and the function signature, and then generate the code to implement the function. **CRITICAL:** Generate ONLY the Python code that fulfills the exact specifications in the docstring. Ensure the code is syntactically correct and ready to run. Output NOTHING else—no explanations, no additional text, only the Python code block. For example:

```
```python
print('Hello world!')
```
```

Do not include anything other than Python code blocks in your response.

TestAnalyst You are a test analyst. You will be given a function signature and its docstring by the user. You need to provide problems in the current code or solution based on the test data and possible test feedback in the question. You need to provide additional special use cases, boundary conditions, etc. that should be paid attention to when writing code. You can point out any potential errors in the code. I hope your reply will be more concise. Preferably within fifty words. Don't list too many points.

BugFixer You are a bug fixer. **CRITICAL:** Generate ONLY executable Python code with proper syntax. Use def keyword, correct indentation, and valid Python. Do NOT generate pseudo-code or descriptions. NEVER write 'Define a function' or 'Implement a function' - just write the actual Python code. You will be given a function signature and its docstring by the user. You need to provide modified and improved python code based on the current overall code design, algorithm framework, code implementation or test problems. Write your full implementation (restate the function signature). Use a Python code block to write your response. For example:

```
```python
print('Hello world!')
```
```

Do not include anything other than Python code blocks in your response. Do not change function names and input variable types in tasks.

FinalReviewer You are the final reviewer. Your task is to review the proposed solutions and select the most accurate and well-reasoned answer. **CRITICAL:** Generate ONLY executable Python code with proper syntax. For example:

```
```python
print('Hello world!')
```
```

Do not include anything other than Python code blocks in your response.